TA14 ACADEMY / CASE STUDY 001

THE HOUSE WON'T COOL

Evidence Before Intervention | Governed HVAC Service Casebook

TECHNICAL TRAINING MANUAL / v0.3 CANDIDATE / 18 OCT 2026

A simulated residential service call that teaches the technician to preserve original reality, investigate an airflow concern, hold refrigerant entry, make a bounded determination, complete authorized work, and prove only the outcome supported by comparable evidence.

CASE ID	SERVICE CONDITION	TRAINING OUTCOME
TA14-ACADEMY-001	Cooling demand not satisfied; indoor 79 F with 72 F setpoint	IMPROVED - bounded, not RESTORED

Governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Training-only boundary

All people, readings, approvals, instrument identifiers and outcomes are fictional. This document is not a field authorization, charging guide, equipment-specific specification, trade credential, or substitute for qualified supervision. Safety and applicable law override training sequence.

Student mission

Before reading the instructor findings, identify the next justified test, decide whether refrigerant-side entry is appropriate, state what remains unknown, and determine which post-intervention claims are supported.

01 / The service call

The customer says the home has been uncomfortable for several hours. The thermostat calls for cooling continuously. The technician must separate the complaint from direct observation before altering equipment.

FIELD	SIMULATED INITIAL RECORD
Indoor thermostat	Cooling mode; 72 F setpoint; 79 F displayed
Outdoor conditions	92 F dry-bulb, afternoon; weather source and time must be recorded
Equipment	Residential split-system cooling unit; make, model, serial and nameplate details intentionally pending
Occupancy / envelope	Windows reported closed; occupancy and recent door activity require confirmation
Stabilization	System observed running for 20 minutes; suitability of interval not yet established
Authority	Customer has requested diagnosis; this does not authorize every intervention

Arrival decision

Observe immediate hazards first. If a dangerous condition exists, make the equipment safe as required and record why original operating conditions could not be fully preserved. Otherwise, avoid resets, thermostat changes, filter removal or gauge connection before the initial record is captured.

Student checkpoint A

Which statements are reports and which are measurements? What additional identity and context information must be collected before comparing this call with another system?

02 / Build the original-state evidence ledger

Use the same identified instrument, measurement point and method for later comparison wherever practicable. A number without its source, operating context and uncertainty is not a complete evidence object.

OBSERVATION	RAW VALUE	LIMITATION
Return / supply dry-bulb	79 F / 66 F	13 F air-side difference; not cooling capacity or charge
Return filter	Visibly heavily loaded	Photo, location and filter specification required
Total external static pressure	0.92 in. w.c.	Tap locations and manufacturer limit unverified
Indoor blower	Operating	Airflow quantity not measured
Outdoor fan / compressor	Observed operating	Does not establish charge, efficiency or capacity
Coil condition	Not directly observed	Do not infer coil cleanliness or frost state

Provenance fields required for every measurement

Case/session ID; technician and equipment identity; date/time and time zone; point location; instrument ID, range, resolution and calibration status; units; operating mode; fan setting; measurement method; raw reading; uncertainty or known limitation; photographs where relevant. Do not fabricate missing calibration or manufacturer data.

Instructor caution

A high-looking static-pressure number may be concerning, but the applicable equipment limit is unknown. A 13 F temperature difference may be worth investigating, but by itself cannot distinguish airflow restriction, load, charge, instrumentation or other causes.

03 / Examine hypotheses before acting

HYPOTHESIS	EVIDENCE FOR	WHAT IS NOT ESTABLISHED
Restricted return path / loaded filter	Visible loaded filter; elevated measured pressure	Actual airflow; exclusive root cause; equipment limit
Refrigerant undercharge	Weak comfort and 13 F temperature difference may prompt investigation	No charge measurement, verified airflow or charge diagnosis
Coil / duct / controls problem	Possible competing explanations	No direct supporting observation in this record

NIRET - Non-Invasive Refrigerant Entry Threshold

Disposition: HOLD refrigerant-side entry. Air-side uncertainty remains material. This case does not establish the current Academy NIRET criteria as satisfied. Do not attach gauges or add/remove refrigerant merely because the home is warm. NIRET concerns evidential necessity; separate qualification, safety and authorization govern whether entry is permitted.

Pre-intervention declared diagnostic determination

Observed: heavily loaded return filter and measured 0.92 in. w.c. total external static pressure. Supported concern: return-side restriction may be contributing to poor cooling. Unresolved: airflow quantity, equipment-specific limit, coil condition, refrigerant condition and unique root cause. Proposed bounded action: verify specified filter and obtain approval for replacement, then repeat comparable measurements.

Decision checkpoint B

Would an apparent low temperature differential authorize charging refrigerant? Explain why or why not. What specific new evidence would cause the technician to reconsider the refrigerant boundary?

04 / Authority and controlled intervention

GATE	SIMULATED STATUS	REQUIRED RESPONSE
Admissible Evidence	Sufficient for limited filter concern, not charge diagnosis	Preserve raw baseline and limitations
Applicable Authority	Work order permits specified filter replacement only	Do not expand scope without new approval
Established Standing	Qualified technician assumed for scenario; credentials not independently demonstrated	Verify identity, competence and safety prerequisites
Current context	No material change assumed before intervention	Re-examine if conditions or permissions change

Authorized simulated action

Confirm filter dimensions and performance characteristics against equipment requirements. Verify safe access and required electrical precautions. Remove and preserve photo evidence of the loaded filter; install the correctly specified replacement; document material, time, work performed and any deviations. No refrigerant-side access is performed.

Stop conditions

HOLD or ESCALATE if correct replacement specifications cannot be established, customer approval is absent, safe access is uncertain, conditions change materially, or the proposed action exceeds standing or scope. Never treat a diagnostic hypothesis as automatic execution permission.

05 / Post-intervention performance record

Post-intervention evidence must be preserved separately from the original state. The simulated comparison below assumes the same fan setting and comparable measurement points; the actual measurement metadata remain training placeholders requiring completion.

MEASUREMENT	ORIGINAL	POST	WHAT IT SHOWS
Return dry-bulb	79 F	78 F	Operating context changed by 1 F
Supply dry-bulb	66 F	60 F	Supply temperature decreased
Air-side difference	13 F	18 F	Difference increased by 5 F
Total external static pressure	0.92 in. w.c.	0.61 in. w.c.	Measured reduction of 0.31 in. w.c.
Filter condition	Heavily loaded	New specified filter	Bounded physical intervention completed
Refrigerant access	None	None	NIRET HOLD maintained

Comparability audit

Confirm tap locations, instrument IDs, mode, fan setting, stabilization, return/supply sensor placement, outdoor load and any changes in occupancy or envelope. Preserve raw and any adjusted data separately. If material comparability is not established, narrow the outcome claim or mark it UNRESOLVED.

Why temperature difference is not capacity

Air-side temperature difference alone does not establish total delivered cooling. Airflow, humidity/latent load, system operating conditions and measurement uncertainty matter. The simulated improvement does not prove design airflow, charge condition or satisfaction of the 72 F setpoint.

06 / Proof of Performance and disposition

Training disposition: IMPROVED (bounded). The fictional post-state record indicates lower measured external static pressure and a larger measured return-to-supply temperature difference following the limited filter intervention, subject to measurement comparability. It does not prove complete restoration, persistent comfort, correct charge, or design performance.

CLAIM	SUPPORTED?	REASON
Filter replaced	Yes, within simulated record	Work action documented
Measured static pressure decreased	Conditionally yes	Comparable points and methods must be confirmed
Measured air-side difference increased	Conditionally yes	Return condition changed; comparison context required
Design airflow restored	No	Airflow not measured or benchmarked
Refrigerant charge correct	No	No justified refrigerant-side evaluation
Home reached 72 F	No	No extended comfort trend
System fully restored	No	Insufficient evidence

What would justify a stronger conclusion?

Document equipment identity and manufacturer criteria; measure or otherwise establish airflow by an appropriate approved method; verify safe operation and relevant coil/duct conditions; preserve comparable post-state observations; and obtain an adequate occupied-space temperature trend. A new scope or refrigerant investigation requires fresh evidence and authority.

07 / Student exercise and instructor key

1. Why not connect gauges immediately?

Instructor reference: Because NIRET necessity is unestablished while material air-side confounders remain, and entry requires separate authorization.

2. What must be frozen before replacing the filter?

Instructor reference: Original operating context, raw measurements, instrument provenance, photos, uncertainty, and declared diagnostic determination.

3. Does the filter appearance prove the sole cause?

Instructor reference: No. It supports a restriction concern, not unique root cause or measured airflow.

4. What is the supported outcome?

Instructor reference: IMPROVED (bounded), subject to comparability; not RESTORED.

5. What invalidates the result?

Instructor reference: Unrecorded changes to measurement points, fan settings, instruments, conditions or baseline; unsupported claims or backfilled evidence.

Critical failure triggers

Premature invasive refrigerant entry; intervention before original-state preservation; fabricated or rewritten measurements; determination backfilled after work; unapproved scope expansion; unsafe practice; or restoration claimed without a comparable post-state record.

08 / Image and showroom production contract

IMAGE	TEACHING FRAME
01	Arrival: reported complaint versus independently observed reality
02	Evidence ledger: equipment, instruments, timestamps and measurement points
03	Original state: loaded filter, temperature difference and static pressure
04	Competing hypotheses: what the data supports and cannot support
05	NIRET HOLD: why gauges are not the next justified step
06	Declared determination and authority boundary
07	Controlled filter intervention and change record
08	Post-state: comparable measurements and retained raw values
09	Proof of Performance: IMPROVED versus RESTORED
10	Learner decision challenge and case summary

Samantha narration rule

Every image gets one objective, a short scene explanation, one decision question, and a transition to the next frame. Narration may explain simulated facts but must never invent readings, approvals, certification or manufacturer criteria.

Publication gates - still open

1. Reconcile the exact 14-step field sequence against the controlling Academy field-lab source. 2. Verify current operative NIRET criteria. 3. Complete equipment/instrument metadata for the simulation. 4. Obtain qualified technical review of the test setup and interpretation. 5. Check all images, narration and examination answers against this record. This is a v0.3 candidate, not a released field standard.

Source alignment: TA14 Academy, Evidence Before Intervention HVAC Service Competency & Assessment Blueprint v0.2 (TA14 Exchange repository). All case-specific readings are fictional.